Day 1 – 19th May

Session 1: Neural Networks (Adam Winchester)

- **Concepts Covered**: Layers (Input, Hidden, Output), Activation Functions (ReLU, Sigmoid, TanH), Weights
- **Activity**: Build a paper neural network (decision logic using weights)
- **Outcome**: Understanding how neural networks simulate decision-making

Session 2: Embedding Knowledge into Optimization (Amir H. Gandomi)

- **Focus**: Optimization types — Evolutionary, Swarm, Gradient-Free
- **Real-World Problems**: TSP, Knapsack, Cantilever Beam, Speed Reducer
- **Key Takeaway**: Optimization improves when domain knowledge (math/stats/engineering) is embedded

Day 2 – 20th May

Session 1: Neural Networks in Business (Adam Winchester)

- **Applications**:
  - *Marketing*: Lead scoring, ad targeting
  - *Finance*: Fraud detection (CNN), trading
  - *HR*: CV screening, turnover prediction
  - *Logistics*: Route optimization
- **Network Types**: FNN, CNN, RNN, LSTM, DNN, Transformers
- **Case Study**: Commonwealth Bank – Fraud detection using CNN (30% reduction in false positives)
- **Activity**: Map business problem to NN solution

Session 2: Startup Journey – JobGen.AI (Kush Bhatia)

- **Focus**: Lean Canvas and startup ideation
- **Example**: AI calorie tracker app from image
- **Team Activity**: Designed *CareerSathi* – AI tool for career problem solving

Day 3 – 21st May

Session 1: VR for Language Learning (Dr. Julia Khukalenko)

- **VR Types**: High Immersion (headset), Low Immersion (screen)
- **Tool**: *Varvara* – Language dialogue simulator
- **Effectiveness**: Best for speaking & listening; not ideal for reading/writing
- **Teacher Readiness**: Adoption linked to prior VR use & IT support

Session 2: AI from Research to Industry (Dr. Farshid Keivanian)

- **Key Concepts**: AI vs ML vs DL with analogies (school/class/smart student)
- **Applications**:
  - *Education*: Smart tutors
  - *Healthcare*: Disease detection
  - *Business*: Chatbots, recommendation systems
- **Ethics & Challenges**: Bias, privacy, responsible AI
- **Case Study Activity**: Apply AI tools to practical problems
- **Research Proposals**: AI tutors for cybersecurity (Torrens x NVIDIA)

Day 4 – 22nd May

Session 1: How capital structure influences profitability of distressed firms in financial crisis

By- Arindam Masanta

Cloud Native development focuses on building and running scalable applications in dynamic environments like public, private, and hybrid clouds. It relies on several core principles:

Microservices architecture breaks applications into small, independent services that can be developed, deployed, and scaled separately. This improves agility, fault isolation, and ease of maintenance.

Containers, especially with Docker, package applications with all their dependencies, ensuring consistent performance across environments. They are lightweight, portable, and fast to start, making them ideal for microservice deployment.

Continuous Integration (CI) automates the building and testing of code whenever changes are made, enabling early detection of bugs and smooth team collaboration.

Continuous Delivery (CD) takes this a step further by automating the release of code to production after passing tests, enabling faster and more reliable software updates.

Orchestration, with tools like Kubernetes, automates the deployment, scaling, and management of containerized applications. It ensures efficient resource usage, high availability, and resilience through features like auto-scaling and self-healing.

Together, these principles create a powerful ecosystem for building modern, efficient, and cloud-ready applications.

Session 2 : Introduction to Optimisation

By – Anita Moghaddam

Optimization is the science of finding the best solution under given constraints. It aims to improve performance, reduce costs, and make better decisions. Mathematically, it involves maximizing or minimizing an objective function while satisfying constraints.

In daily life, optimization appears in route planning, budgeting, and time management. In real-world applications, it's used for modeling systems, making predictions, and controlling processes. Key fields include manufacturing (efficiency), robotics (path planning), and biomedical engineering (treatment design).

Overall, optimization plays a crucial role in solving complex problems across technology and everyday life.

Day 5 – 23rd May

Session 1: Cybersecurity Careers and the Impact of AI & IOT: Shaping the future of Digital Defence (navigating new opportunities and challenges)

By: Joe Cozzupolli

Shaping the Future of Digital Defence

AI and IoT are transforming the cybersecurity landscape, creating smarter systems while expanding the scope of digital threats.

AI introduces a **paradigm shift**—automating threat detection, response, and prediction with unmatched speed and accuracy.

IoT connects billions of devices, improving efficiency but also **widening the attack surface**, making cybersecurity more complex.

AI comes to the rescue by securing vast IoT ecosystems through real-time analysis and autonomous protection.

As technology evolves, **new cybersecurity roles** are emerging: AI Security Analysts, IoT Risk Experts, and Threat Automation Engineers.

The key to success lies in **human–machine collaboration**, blending strategic thinking with AI's speed and precision.

Together, AI and IoT are reshaping careers and redefining how we protect the digital world.

Day 5 – 23rd May

Session 2: Internet of Things (IoT)

By: Florence Zhao

The **Internet of Things (IoT)** connects everyday objects to the internet, enabling real-time data sharing and automation.

- **Why It Matters**: Boosts efficiency, reduces costs, and enhances decision-making.
- **Definition**: A network of smart devices exchanging data.
- **Example**: Smart meters track energy use in real time.
- **In Mining**: Improves safety, monitors equipment, and predicts failures.
- **Types**: Consumer, Industrial, and Infrastructure IoT.
- **Value**: Powers innovation and digital transformation across industries.

IoT is making the world smarter, safer, and more connected.

Day 6 – 24th May

Session 1: From Tool to Assistant (the transformation of information technology through generative AI)

By: Roger Lawrence

IT is evolving—from a static tool to a dynamic assistant—driven by the rise of **Generative AI**. This shift enables systems to understand, generate, reason, and act alongside humans.

- **AI Assistance**: IT now collaborates, understands intent, and provides intelligent support.
- **RAG (Retrieval-Augmented Generation)**: Enhances AI with real-time, accurate knowledge retrieval.
- **Diffusion Models**: Fuel content creation in **image, audio, and video** generation.
- **Inference & Training**: Enables AI to **reason**, learn, and adapt to new data.
- **Prompt Engineering**: The new skill of designing effective AI inputs for powerful outcomes.
- **AI Agents**: Autonomous tools that complete tasks, make decisions, and adapt dynamically.
- **AI Operators**: Emerging roles focused on deploying and managing AI systems.
- **Vibe Coding**: Human-AI coding collaboration through natural language and co-creation.
- **Model Context Protocol**: Maintains context and memory for coherent, long-term AI interaction.

Generative AI is reshaping IT into an **intelligent, creative, and interactive assistant**—marking a fundamental shift in how we build and use technology.

Day 6 – 24th May

Session 2: Introduction to AI, ML & Data Science

By: Duc-Minh Pham

AI enables machines to mimic human intelligence.
ML is a subset of AI that learns from data.
Data Science uses AI/ML + statistics to extract insights.

- **Relationships**: ML is part of AI; both overlap with Data Science.
- **Key Techniques**: Prediction, classification, data analysis, visualization.
- **Learning Types**:
  - **Supervised** – learns from labeled data
  - **Reinforcement** – learns via feedback/rewards
- **Applications**: Healthcare, finance, robotics, recommendations.
- **Core Methods**: Neural networks, decision trees, regression.

Together, AI, ML, and Data Science power intelligent, data-driven solutions.

Day 7 – 25th May

Session 1: Quantum Computing vs Classical Physics

By: Tanveer Molla

Classical computing is **deterministic** and uses bits (0 or 1).
Quantum computing uses **qubits**—in **superposition**, enabling powerful, parallel computation.

- **Quantum Physics**: Probabilistic, strange measurements, uncertainty.
- **Applications**: Prime factorization, secure communication.
- **Qubit Platforms**: Donor qubits, superconductors, trapped ions.
- **Quantum Supremacy**: Quantum outperforms classical in specific tasks.

Quantum computing taps into the weirdness of quantum mechanics to solve problems classical computers can't.

Day 7 – 25th May

Session 2: Spatial Computing & AR Evolution

By: David Francis

Spatial computing blends digital and physical worlds, enabling interaction through AR, VR, and AI.

- **Early AR**: Tools like DAQRI led the way but faced missteps and pivots.
- **First Opportunities**: Focused on industrial and enterprise use.
- **Wearable Computing**: Moved to the edge—hands-free, smart, context-aware.
- **B2C Expansion**: AR reached consumers via phones and glasses.
- **Next Era**: AI-enhanced wearables bring smarter, immersive experiences.

Spatial computing is grounded in reality, driven by AI, and shaping the future of human-tech interaction.

Day 8 – 26th May

Session 1: Intro to AR/VR & Unity Interaction Toolkit

By: Matt Cabanag

AR/VR create immersive digital experiences blending virtual and real worlds.

🔷 **Unity**: A leading platform for building AR/VR apps.

🔷 **EZPZ Interaction Toolkit**: Simplifies development with ready-to-use tools for grabbing, touching, and UI interaction in VR.

🔷 **Key Features**:

• Drag-and-drop components
• Hand tracking support
• Cross-platform compatibility

🔷 **Basic Unity Functions**:

• `Start()` and `Update()` – control behavior
• `Transform` – move, rotate, scale objects
• Physics and event triggers for interaction

Unity + EZPZ Toolkit makes AR/VR development more accessible and fast for creators.

Day 8 – 26th May

Session 2: Security for AI & AI for Security

By: Vik Verma

AI enhances cybersecurity but brings new risks. Securing AI and leveraging AI for defense are crucial.

- **Frameworks**: NIST and Zero Trust guide AI/ML security with strict access controls.
- **AI Types**: Traditional, agentic (autonomous), and custom models.
- **Securing AI**: Protect data, models, and deployment.
- **AI for Security**: Threat detection, investigations, and AI copilots aiding analysts.

Balancing AI’s power and protection is key to future cybersecurity.

Day 9 – 27th May

Session 1: Big Data & Business Analytics

By: Saradhi Motamarri

Big Data is growing explosively, driven by the **7 V's**: Volume, Velocity, Variety, Veracity, Value, Variability, and Visualization.

🔷 **Business Analytics** uses data to improve decisions through:

- **Descriptive** (what happened)
- **Predictive** (what will happen)
- **Prescriptive** (what to do)

🔷 **Skills in Demand**: Data analysis, visualization, statistical modeling, and BI tools.

🔷 **Top BI Software (2021)**: Leaders in market share driving analytics adoption.

🔷 **Trends (2023)**: AI integration, real-time analytics, cloud BI.

🔷 **Global Use**: Countries leading in big data analytics adoption.

Big Data & Business Analytics empower smarter, data-driven business strategies.

Day 9 – 27th May

Session 2: Spatial Computing & Enterprise Innovation

By: David Francis

Spatial computing merges physical and digital worlds, transforming how engineers and workers interact with data.

🔷 **Key Elements**:

- **Digital Twins** & 3D models for real-world simulation
- **Wearables** & connected workers at the edge
- **Egocentric AI** enabling intuitive, user-focused tools
- **VPS Systems** for spatial positioning in enterprises

🔷 **Opportunities**:

- Bridging the **knowledge gap**
- **Democratizing engineering** through immersive tech
- Enabling **new collaboration paradigms**

🔷 **Challenges**:

- **Planning pain** (e.g., fixing vs flexibility)
- Risks in **digitization** and data overload

Spatial computing redefines industrial workflows—smarter, immersive, and human-centered.

Day 10 – 28th May

Session 1: Reinforcement Learning (RL) Overview

By: Dr. Mohammad B. Naghibi S.

Reinforcement Learning (RL) is a branch of AI/ML where agents learn by interacting with their environment—**trial, error, reward, and punishment**.

🔷 **AI → ML → RL**:

RL is distinct from **supervised** (labeled data) and **unsupervised learning** (hidden patterns); it learns from **feedback over time**.

🔷 **Key Concepts**:

- **Agent-Environment Interface**
- **Rewards**, **Returns** (short/long-term)
- **Value Function**
- **Bellman Equation**
- **Markov Decision Process (MDP)**

🔷 **RL Algorithms**:

- **Dynamic Programming**
- **Monte Carlo Methods**
- **Temporal Difference Learning**
- **Generalized Policy Iteration**

RL mirrors human/animal learning—adapting behavior to maximize future rewards.

Day 10 – 28th May

Session 2: Oracle ERP Cloud: AI-Powered Enterprise Applications

By: Arindam Masanta

Oracle ERP Cloud integrates AI across enterprise functions, driving **efficiency, innovation, and productivity**.

🔷 Oracle Cloud Applications

- **ERP** – Financial excellence with AI-driven insights
- **HCM** – Smart workforce management
- **SCM** – Efficient, AI-optimized supply chains
- **Retail** – Personalized, data-driven customer experience
- **Education** – Adaptive, AI-enhanced learning

🔷 AI Transformation

- Rise of **AI agents** powering automation & decision-making
- Boosts productivity across finance, HR, and operations
- **CFO-focused AI tools** support smarter forecasting and planning

Oracle's AI-infused SaaS ecosystem helps enterprises **achieve more with less** across sectors.